

HIGH-THROUGHPUT NONLINEAR CLOSED-ORBIT CALCULATION AND RESPONSE-ERROR ANALYSIS IN THE LATEST CESR MODEL BASED ON SCIBMAD

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Abstract

We present a batched RF-on closed-orbit calculation in SciBmad for the latest validated CESR charged-particle lattice. A nominal GTPSA response initializes frozen-Jacobian Newton iterations with explicit closure checks and an automatic-differentiation fallback. A 36001-state amplitude sweep over all, horizontal, and vertical steering subspaces converges 36000 states; 62 of 63 fallback attempts recover, and every converged state closes below 10^{-13} . At the first sampled radius where the all-control mean residual exceeds $1\mu\text{m}$, $\rho = 1.13$, the means are $1.445\mu\text{m}$ in x and $1.153\mu\text{m}$ in y . Exact complete-element sourcing over 1177 elements reconstructs the horizontal and vertical quadratic targets to relative residuals of 1.309×10^{-14} and 1.312×10^{-14} . Normal sextupoles supply 96.508% and 98.647% of the corresponding signed projections. A matched 9001-state comparison requires 17.989 s for 16-thread SciBmad physics and 136.797 s for Bmad/Tao configured with 16 OpenMP threads, a 7.605-fold wall-clock speedup; including reusable-model SciBmad setup requires 22.842 s.

INTRODUCTION

Accelerator datasets intended for model calibration, surrogate learning, and control studies must be more than large and computationally inexpensive: their labels should represent physically self-consistent states, carry reproducible solver provenance, and remain accurate over a quantified input domain. High-speed differentiable simulation makes large-scale generation practical [1–3], and learned map models show the value of such data for beam-based error identification [4]. For an orbit dataset, the central label is the closed orbit associated with each sampled control state. It should satisfy the six-dimensional RF-on periodic condition, and the error from replacing that nonlinear solution with a nominal linear response should be known.

The orbit-response matrix provides an efficient local model, but nonlinear lattice maps have visible effects with large corrector amplitudes, particularly through sextupoles. Large-amplitude and off-energy closed-orbit measurements have therefore been used to characterize and correct nonlinear optics [5, 6]. Generating reference data in this regime requires repeated nonlinear fixed-point solutions. A useful implementation must combine throughput with explicit closure tests, robust convergence, and a mapped relationship between linear-response error and nonlinear-solver survival.

We apply this workflow with SciBmad [8] to the RF-on main branch of a Bmad-compatible latest CESR model [7,

9]: a 6 GeV positron lattice with four 1.5 MV RF cavities, 103 independently varied normal-steering controls, and 144 detector locations reporting both transverse planes. The selected controls comprise 58 horizontal and 45 vertical channels; the detector vector therefore contains 288 ordered observables. We map response error and convergence over corrector amplitude and input subspace, then attribute the quadratic detector target to complete lattice elements. The calculation does not vary girder pitch or photon branches, so the documented curved-DQX pitch and general mirror-optics limitations do not enter the reported observables. The study is synthetic numerical analysis, not a real-machine validation.

ORBIT CALCULATION METHODOLOGY

The CESR configuration is loaded from `latest_cesr_scibmad_repaired.jl` in SciBmad 0.4.1 with RF on and branch 0. All dimensions and ordered labels are discovered from the runtime control and detector registries rather than fixed historical constants.

For corrector vector k , the RF-on closed orbit z_* satisfies the six-dimensional fixed-point equation

$$F(z_*, k) = \mathcal{M}(z_*, k) - z_* = 0. \quad (1)$$

SciBmad evaluates many corrector states as one batch. A cached 6×103 nominal response supplies the initial value as $z_{\text{init}} = z_0 + R\Delta k$. Newton iterations then evaluate the nonlinear one-turn residual while reusing one nominal 6×6 Jacobian factorization. When a lane does not converge, a sample-specific automatic-differentiation Jacobian is used as fallback. Every accepted state is tested against the six-dimensional periodic closure, with relative and absolute tolerances of 10^{-12} and 10^{-13} .

LATEST-CESR NONLINEAR CALCULATION

For the active correctors A in a selected input subspace, let $k_{\text{rms}} = \text{rms}(\Delta k_A)$ and define $\rho = k_{\text{rms}}/5\mu\text{rad}$. Each sampled Gaussian direction d is normalized independently to $\text{rms}(d_A) = 1$, so that $\Delta k = \rho d \cdot 5\mu\text{rad}$. For all-, horizontal-, and vertical-control inputs, 600 fixed directions are reused at 20 positive radii from $\rho = 0.1$ to 64. Together with one shared nominal reference, this gives 36001 unique exact states. The five radii retained from the earlier matched table, $\rho = 1.13, 3.2, 4.53, 6.4$, and 9.05 , contribute 9000 nonzero states and are all complete in the latest-ring run.

Table 1 summarizes the matched latest-ring calculation. One all-control lane at $\rho = 51.2$ does not converge; all other sampled cells, including $\rho = 64$, are complete. This isolated non-monotonic failure is reported as solver-survival evidence, not as a physical aperture boundary. For the matched 9001-state comparison, both codes return all states. SciBmad uses no fallback, requires at most seven iterations, and has maximum closure 9.992×10^{-11} . Tao uses one persistent process configured with 16 OpenMP threads and evaluates the input states sequentially.

Table 1: Latest-CESR matched nonlinear closed-orbit calculation.

Calculation	States	Time [s]	Rate [s^{-1}]
Bmad/Tao (16 thr.)	9000	136.797	65.791
SciBmad physics (16 thr.)	9000	17.989	500.318
SciBmad incl. setup	9000	22.842	394.018

ORBIT RESPONSE ERROR ANALYSIS

General amplitude dependence

For detector-orbit vector u , we can expand about the nominal corrector state input,

$$u(\Delta k) = u_0 + J\Delta k + \frac{1}{2}H[\Delta k, \Delta k] + \frac{1}{6}T[\Delta k, \Delta k, \Delta k] + \dots \quad (2)$$

The nominal linear-response residual is $e = u(\Delta k) - u_0 - J\Delta k$. The general sweep uses the same 600 directions at 20 radii from $\rho = 0.1$ to 64 in each input subspace, which is 36001 nonlinear states in total including the nominal reference. Detector rms residuals are summarized by the mean and largest sampled direction at each radius in Fig. 1.

At $\rho = 1.13$, the all-control mean residuals are $1.445 \mu\text{m}$ in x and $1.153 \mu\text{m}$ in y . Vertical-only inputs produce $1.053 \mu\text{m}$ in x but only $0.008259 \mu\text{m}$ in y ; horizontal-only inputs produce $0.7308 \mu\text{m}$ in x and a y residual below $2 \times 10^{-13} \mu\text{m}$. Thus $\rho = 1.13$ is the first sampled all-control radius at which the mean in either output plane exceeds an illustrative $1 \mu\text{m}$ label-error budget. This is a discrete sampled crossing, not an interpolated machine limit.

The principal-plane channels remain approximately quadratic through the moderate-amplitude range. The much smaller vertical-only y residual grows approximately cubically over the five tabulated radii, while the horizontal-only y channel remains at numerical-zero scale. A dataset must therefore state its accepted output error and cannot infer one global input limit from corrector rms alone.

All 600 directions in all three scenarios converge through $\rho = 36.2$. At $\rho = 51.2$, 599/600 all-control directions converge; every other sampled scenario-radius cell is complete. Overall, 36000/36001 states converge, with 62 lanes recovered by fallback and one retained failure whose closure norm is 4.35066.

Second-order lattice-element attribution

The amplitude sweep identifies the quadratic regime but does not assign its detector error to lattice elements. For

the 100 corrector-direction pairs used below, the implicit GTPSA solution supplies the first- and second-order periodic orbit at every element boundary.

Let A_j be the first-order (orbit) map of complete element j , and let $S_{j,-}^{(d)}$ and $S_{j,+}^{(d)}$ be the direction-contracted second-order states at its entrance and exit. The new local source generated by that element will then be

$$g_j^{(d)} = S_{j,+}^{(d)} - A_j S_{j,-}^{(d)}. \quad (3)$$

Each source is propagated through the periodic ring to all 144 detectors. For output plane $p \in \{x, y\}$, let $Q_p^{(d)}$ denote the all-corrector direction-contracted quadratic target. Summing propagated vectors from all 1177 elements reconstructs $Q_x^{(d)}$ and $Q_y^{(d)}$, giving ensemble relative residuals 1.309×10^{-14} and 1.312×10^{-14} , respectively.

For family F , let $C_{F,p}^{(d)}$ denote its summed propagated detector vector. Then we can calculate both the ensemble signed projection and magnitude ratio

$$\eta_{F,p} = \frac{\sum_d C_{F,p}^{(d)} \cdot Q_p^{(d)}}{\sum_d \|Q_p^{(d)}\|^2}, \quad \mu_{F,p} = \left(\frac{\sum_d \|C_{F,p}^{(d)}\|^2}{\sum_d \|Q_p^{(d)}\|^2} \right)^{1/2}. \quad (4)$$

First-order orbit-response error versus corrector input scale

CESR RF-on; gtpsa response; 600 directions/scale; rho = 1: 5 microrad active-control RMS

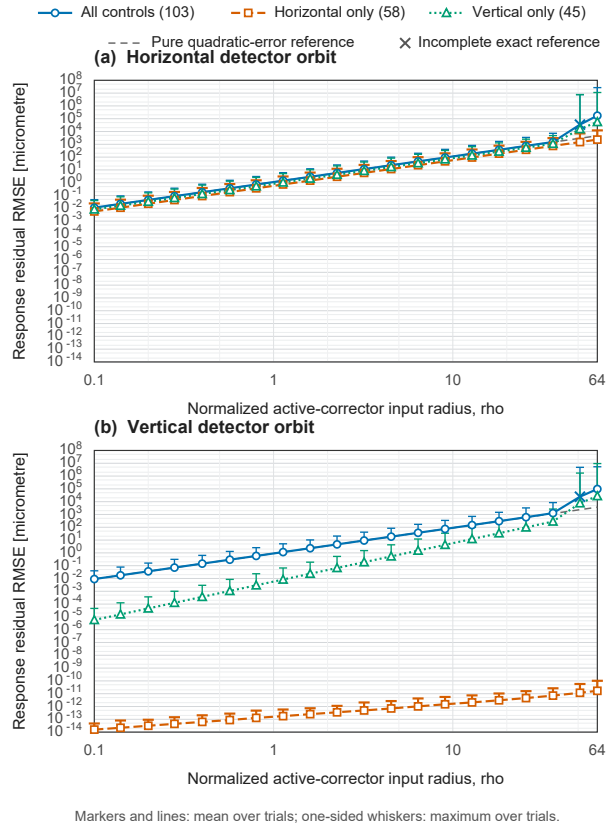


Figure 1: Growth of the horizontal and vertical nonlinear orbit-response residuals with corrector-input radius over 600 fixed directions per radius. Curves show the mean detector rms, whiskers the largest sampled direction, and crosses cells with incomplete exact references.

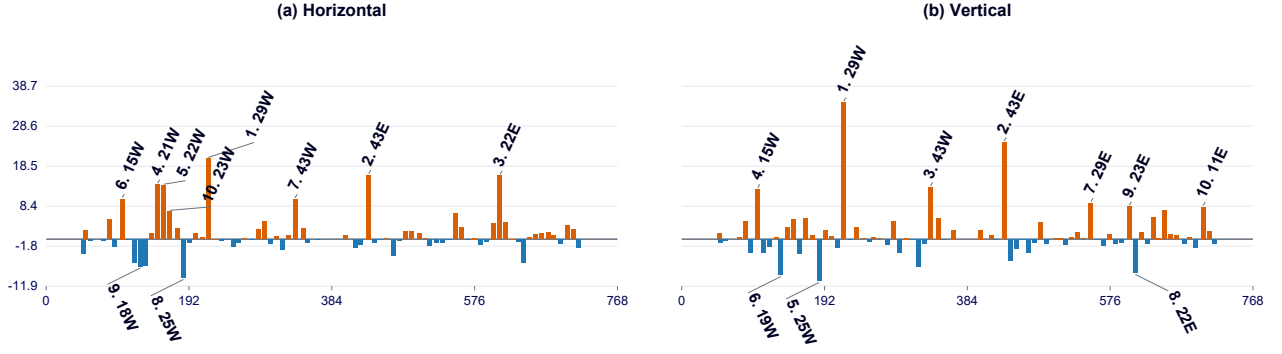


Figure 2: Signed complete-element projections of the 76 active normal sextupoles onto the all-corrector quadratic detector targets, plotted against ring position for (a) horizontal and (b) vertical output. Labels mark the ten largest absolute projections in each plane; positive and negative bars display constructive and cancelling contributions to the family sum.

Family	N	Horizontal Q_x		Vertical Q_y	
		$\eta_{F,x}$ [%]	$\mu_{F,x}$ [%]	$\eta_{F,y}$ [%]	$\mu_{F,y}$ [%]
Normal sextupole	76	+96.508	99.295	+98.647	99.859
Sector bend	98	+1.702	11.993	+0.384	6.833
Drift/geometric	562	+1.008	6.843	+0.502	5.128
Kicker	82	+0.395	2.480	+0.246	1.966
Quadrupole	122	+0.357	2.349	+0.203	1.830
RF cavity	4	+0.021	0.146	+0.017	0.135
Wiggler	3	+0.005	0.067	+0.001	0.074
Octupole	2	+0.003	0.028	-0.000	0.025
Other sextupole	6	+0.002	0.015	+0.000	0.009
Marker	222	+0.000	0.000	-0.000	0.000

Table 2: Complete-element family attribution of the horizontal and vertical all-corrector quadratic targets. Signed projections $\eta_{F,p}$ are additive; magnitude ratios $\mu_{F,p}$ are not.

For either output plane the $\eta_{F,p}$ values add to unity; the $\mu_{F,p}$ values would not as the vectors have different directions. The complete 10-family partition is shown in Table 2. Normal sextupoles supply 96.508% of the horizontal and 98.647% of the vertical signed projection. Sector bends and drift/geometric maps are the leading secondary families. Kicker and quadrupole sources remain visible below the percent level, while wiggler and RF sources are negligible for these observables. Coupling, fringe, and kinematic effects remain inside the complete-element maps.

The complete normal-sextupole vectors leave relative residuals of 23.618% for x and 15.569% for y , whereas the all-element reconstruction retains the near-machine-precision closure quoted above. The sextupole family is therefore the dominant projected source but not a closed decomposition by itself; the remaining complete-element families and their interference must be retained when reconstructing the detector vector.

Figure 2 resolves the normal-sextupole family into its 76 signed element projections around the 768.438 m ring. The positive and negative bars show that the family result is a signed, phase-dependent sum rather than a collection of positive error shares.

Direction dependence and physical interpretation

Normal-sextupole dominance is robust across the fixed direction ensemble, but individual directions exhibit substan-

tial constructive and cancelling interference. Table 3 gives the P10, median, and P90 direction-resolved family statistics. The median signed projections are 96.590% in x and 99.775% in y ; the P90 signed values exceed 100%, which is allowed because the remaining family vectors cancel part of the normal-sextupole vector.

This behavior is consistent with the ordinary normal-sextupole local source and periodic closed-orbit transport picture [10, 11], but the present exact attribution does not reduce the complete coupled transport to an uncoupled thin-lens predictor. A direction-matched source-beta-phase correlation study has not yet been rerun in production on the latest lattice; historical-ring correlations are therefore not used as latest-CESR evidence.

Table 3: Direction-resolved normal-sextupole family statistics [%].

Statistic	$\eta_{F,x}$	$\mu_{F,x}$	$\eta_{F,y}$	$\mu_{F,y}$
P10	90.452	95.236	89.809	92.310
Median	96.590	99.178	99.775	99.875
P90	101.268	103.487	108.184	110.876

CONCLUSION

Batched latest-CESR SciBmad calculations converged for 36000 of 36001 sampled RF-on states, including 62 recovered fallback lanes, with accepted closure below 10^{-13} . The amplitude sweep confirms that linear-response error depends strongly on both corrector subspace and detector channel, so a single corrector-rms limit is insufficient for dataset sampling. Exact sourcing over all 1177 elements closes both quadratic detector targets near machine precision and identifies the 76 active normal sextupoles as the leading signed source, at 96.508% in x and 98.647% in y . These latest-lattice results support nonlinear label generation with explicit closure, fallback, and channel-aware sampling metadata. The matched 9001-state calculation gives a 7.605-fold SciBmad physics-time wall-clock speedup over 16-thread native Bmad/Tao; including reusable-model SciBmad setup, the speedup is 5.989-fold.

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